

Kannika Armstrong (Jaikham)

253-282-8828 | akannika80@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | Puyallup, WA

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, PostgreSQL, MySQL, HTML, CSS, PHP, R, Rust
Frameworks: TensorFlow, Streamlit, Bootstrap, Node.js, JUnit, WordPress, FastAPI
Libraries: scikit-learn, Pandas, NumPy, Matplotlib, ReportLab, Plotly, Ollama
Tools: SQL Server, TiDB, Git, GitHub, Jupyter Notebook, GitHub Actions, VS Code, PyCharm, IntelliJ

EDUCATION

University of Washington Tacoma Tacoma, WA
Master of Science in Computer Science and Systems Expected June 2026
• Relevant coursework: Advanced Algorithms, Machine Learning, Applied Distributed Computing, Cryptography, Database Systems Internals
Bachelor of Science in Computer Science and Systems Jan 2021 - Dec 2022
Chiang Mai University Chiang Mai, Thailand
Bachelor of Science in Physics (Minor in Mathematics) May 1999 - Mar 2004

TECHNICAL PROJECTS & RESEARCH

Portfolio Website (for additional information and projects) | a-kannika.github.io/KannikaA/

Machine Learning for Social Media User Profiling | [GitHub](#)

- Improved gender prediction accuracy to 0.76 (vs. 0.59 baseline), achieving a +28.8% in model performance.
- Increased age prediction accuracy to 0.63 (vs. 0.59 baseline) by selecting and tuning an optimal SVM classifier.
- Outperformed the baseline RMSE for 3 of 5 personality traits (Neuroticism, Extraversion, Conscientiousness) by implementing a Gradient Boosting Regressor.
- Trained and benchmarked a suite of models (including SVM, Logistic Regression, GBR, and BERT w/ LSTM) on a dataset of 9,500 user profiles using scikit-learn and Pandas.

Retrieval Augmented Generation (RAG) with TiDB Vector Search | [GitHub](#)

- Orchestrated a RAG pipeline leveraging TiDB's native vector search with cosine similarity for high-accuracy semantic document retrieval.
- Integrated BAAI/bge-m3 transformer (1024-dim) for document embedding and Llama 3.2 via Ollama for context-aware question answering.
- Constructed an end-to-end data pipeline: from embedding generation and TiDB storage to vector similarity search and LLM response synthesis.

Empirical Analysis of Network Flow Algorithms | [GitHub](#)

- Directed a 4-developer team to implement and analyze the performance of complex network flow algorithms.
- Implemented multiple max-flow algorithms in Java: Ford-Fulkerson, Scaling Ford-Fulkerson, and Preflow-Push.
- Engineered a system to parse graph data from files and dynamically manage residual graphs during execution.
- Benchmarked algorithm runtimes against diverse graph structures to identify and document performance trade-offs.

Master's Thesis | Optimizing Performance and Efficiency of Lattice-Based Post-Quantum Cryptography Algorithms in Resource-Constrained UAV Environments (In Progress)

- Architecting and optimizing a secure UAV protocol (integrating Kyber PQC and Ascon-MAC) for resource-constrained hardware, with formal verification via ProVerif.

PROFESSIONAL EXPERIENCE

Teaching Assistant & Academic Mentor Sep 2024 – Jun 2025 (UW); 2020 – 2024 (Pierce)
University of Washington & Pierce College Tacoma & Puyallup, WA

- Led lab sessions, graded Java programming projects, and mentored students in debugging.
- Mentored 50+ students in Algorithms, OOP (Java/Python), and SQL queries.
- Taught collaborative project workflows using Git, GitHub, and VS Code.

Meteorologist & Secretary to the Deputy Director-General May 2006 – May 2018
Thai Meteorological Department Bangkok, Thailand

- Developed predictive models and analyzed large-scale climate data to provide actionable insights.
- Coordinated inter-agency academic projects, advised executive leadership, and managed project lifecycles by reviewing, summarizing, and drafting technical reports and policy.